TITLE: A Novel Method for Rhythmic Imitation of Finger Movements and the Effects of Auditory Stimuli and Feedback

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ABSTRACT (350 words so far, (limit =350))

Manual dexterity, essential for daily and professional activities, relies on accurate and timely coordinated hand and finger movements Rhythmic finger-movement training can be beneficial, and incorporating music can further boost engagement and motivation. Solutions for rhythmic finger training are scarce and effects of auditory stimuli and feedback on performance remain largely unstudied. We developed a method for finger movement reproduction of the rhythmic song structure. The solution comprises a rhythmic imitation platform called Rhythm Karaoke, which synchronizes auditory stimuli with finger taps recorded by a manual dexterity device. It features adaptive sound synthesis that responds to users' tempo and permits multiple auditory stimuli (notes vs. speech, melodic vs. monotone), real-time auditory feedback and rhythmic error quantification.

We hypothesized that melodic and semantic content (speech) would enhance rhythmic precision of finger tapping compared to monotone and non-semantic content (notes). We also investigated whether auditory feedback would enhance performance and how different conditions would affect motivation and preference. Healthy participants (n=30) reproduced rhythmic sequences of three songs in a within-subject design crossing Melody (melodic / monotone), Semantic (speech / notes) and Feedback (present / absent). The rhythm error of reproduced sequences was calculated. The Intrinsic Motivation Inventory was used to assess motivation and subjective experience. Test-retest reliability was studied in the easiest song (melodic note and melodic speech conditions) in a separate group of healthy participants (N=20).

Finger tapping to different songs resulted in significantly different rhythmic error. ANOVA revealed significant effects of Melody and Semantic content as well as their interaction: melody improved performance only for non-semantic note sequences. Auditory feedback only improved performance in the semantic condition, reducing error in note and increasing error in speech condition. Participants preferred melodic conditions, reporting lower perceived tension and higher interest and satisfaction. Reliability was higher in melodic note compared to melodic speech conditions, with excellent test-retest reliability for the melodic note with auditory feedback condition.

Our solution for finger movement reproduction of rhythmic song structures showed benefit of melody and semantic stimuli on rhythmic precision and participant motivation. The method may have potential as a neurorehabilitation tool for manual dexterity training.

1 INTRODUCTION

Manual dexterity can be defined as the ability to perform accurate and coordinated hand and finger movements, as well as relatively independent finger movements, to grasp and manipulate objects [1]. It is essential in many daily and professional activities [37, 38]. Manual dexterity allows performance of various tasks, ranging from daily tasks (e.g. writing, dressing) to music to visual arts [37, 39, 40, 49]. Rhythm and timing are among the key components of manual dexterity alongside force control, finger independence, and motor sequence performance [1]. Rhythmic imitation exercises are widely recognized for their effectiveness in motor and cognitive development [11, 41, 42, 43]. They improve coordination, balance, and promote brain maturation [27]. These exercises are particularly beneficial in motor rehabilitation programs [30, 31, 32], where the ability to reproduce precise rhythmic sequences is fundamental for the recovery of timing of finger movements [1]. Beyond methods allowing rhythmic reproduction of non-melodic notes [71] music-based interventions are of special interest because rhythm and melody are processed by neural systems overlapping with those controlling movement [11, 36], making music a powerful medium for engaging motor and cognitive systems [2, 3, 4]. Rhythm is also intrinsic to speech, which unfolds in quasi-regular temporal patterns across multiple timescales from syllables to prosody [80].

Despite the recognized benefits of rhythmic exercises in motor rehabilitation [72, 1], many traditional approaches rely on simplified tasks such as hand or foot tapping in time with a metronome or rhythmic audio cues [62]. These methods often measure rhythm using global metrics like mean asynchrony or error counts, which can overlook subtle variations in timing and are less reliable in non-expert or clinical populations [14, 46]. As a result, they may fail to capture fine-grained motor timing performance, particularly under more ecologically valid conditions involving musical or speech-based stimuli. Moreover, previous methods rarely compare different types of auditory stimuli, such as melodic versus monotone sounds or speech versus non-speech sounds, even though these factors have been shown to influence sensorimotor coupling and rhythmic engagement [10, 47, 48].

Some studies suggest that temporal predictability and beat salience in music can enhance rhythmic precision and perceived pleasantness during motor tasks [75], and that melodic and familiar auditory patterns can facilitate motor coordination [73]. Such

findings support the potential for speech and music with melodic structure to improve performance and engagement in rhythmic tasks.

The main objective of this study was to develop a rhythmic imitation platform capable of capturing fine-grained timing differences in manual finger movements. A secondary aim was to evaluate the effects of various auditory stimuli and feedback conditions on rhythmic precision, perceived motivation, and measurement reliability. Using a manual dexterity measurement device, participants reproduced heard rhythmic sequences via a platform developed for this study. The sequences varied along two dimensions: Melody (Melodic or Monotone) and Semantic Content (Speech or Piano Notes). The sound stimuli used included monotone notes, melodic notes, monotone speech, and melodic speech. Participants were evaluated on their ability to imitate these different types of sound stimuli, with and without auditory feedback of sounds generated from finger tapping. We hypothesized that melodic stimuli (melodic notes and melodic speech) would improve rhythmic precision because melodies enhance beat salience and predictability [75], while speech stimuli (monotone and melodic speech) would improve rhythm given shared neural pathways between speech and manual motor planning [74] and speech and music perception [76]. We expected that these auditory features would guide motor timing more effectively than monotone non-semantic stimuli. We also hypothesized that auditory feedback would improve precision by reinforcing timing cues and stabilizing performance [19]. Finally, we predicted that participants would prefer melodic conditions over monotone ones, due to increased engagement and reduced tension reported in melodic contexts [4].

2 METHOD

2.1 Participants

We recruited 30 right-handed participants (18 males and 12 females) with an average age of 23.7 years (standard deviation: 4.2 years). Among them, 93.33% were right-handed and 23.3% were musicians. Participants were recruited through announcements at our institution. Each participant provided written informed consent and the study was approved by the Ethics Committee of the Université Paris Cité (CER U-Paris Cité N◦ IRB: IORG0010120-2025-33) and was in accordance with the ethical standards of the 1964 Declaration of Helsinki. The sample size was determined by a statistical power analysis based on the results of a previous study using a similar device [63]. By setting the desired power at 0.80, the Type I error risk at 0.05, and using an estimated effect size of 0.74, this analysis, performed with the Python libraries statsmodels and scipy, estimated that a minimum of 30 participants was necessary to ensure the robustness of the results. The effect size of 0.74 was chosen based on a

previous pilot study we conducted that examined similar effects in a comparable context [50].

2.2 Experimental Design and Procedure

Participants took part in a 30-minute session involving the performance of three different rhythms ("La Vie en Rose" by Édith Piaf, "Papaoutai" by Stromae, and "Bare Necessities" from The Jungle Book, sung by Phil Harris and Bruce Reitherman). Each trial began with a listening phase in which the reference rhythm was presented once in one of four modes (monotone note, melodic note, monotone speech, melodic speech). Participants then reproduced the rhythm either (i) with real-time auditory feedback, where every finger tap triggered the corresponding note or syllable through the headphones, or (ii) without feedback, in which taps were silent (see Section 2.3.3 “Sound Synthesis” for implementation details). During reproduction a horizontal status bar filled on the screen to indicate trial progress.

2.2.1 Familiarization Phase

Participants began by practicing a rhythm different from those used in the main experiment, specifically "Emmenez-moi" by Charles Aznavour. Each practice trial followed the same imitation paradigm as in the main study: the reference rhythm was played once, then participants reproduced it. They practiced as much as they wanted on the "note" and "speech" conditions with auditory feedback, allowing familiarization with the various modalities of the experiment. To prepare participants for the absence of auditory feedback, a final "speech" condition without auditory feedback was included. This step aimed to avoid any surprise due to the lack of auditory feedback during the main tasks. Participants could re-listen to the example and restart the exercise as many times as needed. This familiarization phase ensured that participants fully understood the requirements of each experimental condition and were comfortable with the different configurations [52]. This prior practice minimized confounding variables related to learning and ensured more stable performances during the main tasks. The familiarization phase lasted approximately 8 minutes.

2.2.2 Experimental Phase and Bias Control

The experiment included 24 configurations (3 rhythms * 4 conditions * 2 auditory feedback), each repeated 3 times, totalling 72 trials (see Fig. 1). The configurations were pseudo-randomized to avoid the repetition of the same conditions or rhythms consecutively. This randomization aimed to control potential biases and ensure a balanced distribution of tasks [53], aimed to minimize learning or training effects by distributing the configurations evenly throughout the experiment [53].

Breaks were offered between each block to prevent fatigue and maintain a consistent performance level throughout the experiment [53].

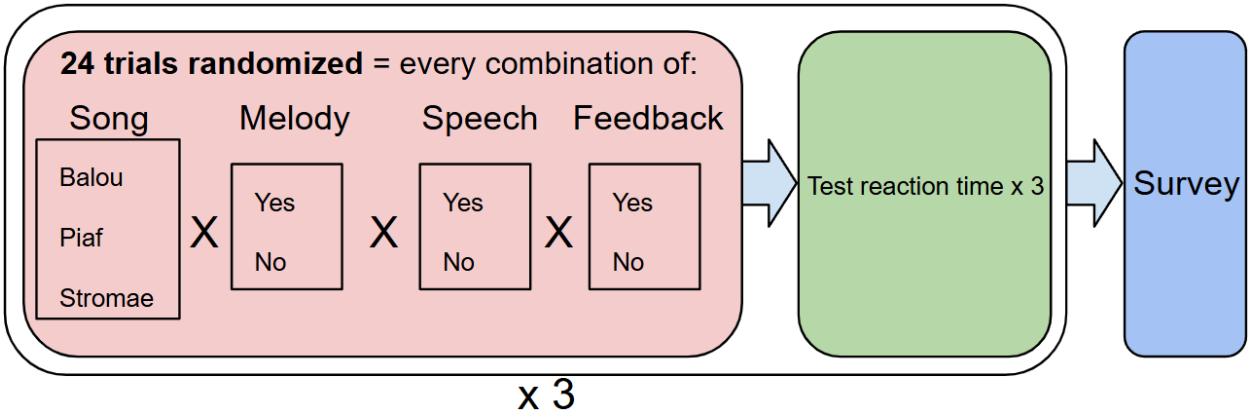


Fig. 1. Experimental flow and counter-balancing scheme. Each block comprised 24 pseudo-randomised trial configurations (3 songs × 2 melodic modes × 2 speech modes × 2 feedback modes). The entire sequence was repeated three times, with a reaction-time test after every block and a final preference survey.

2.2.3 Final Questionnaire

At the end of the session, participants completed a questionnaire based on the Intrinsic Motivation Inventory [51]. For each condition and auditory feedback type, they rated perceived tension, interest, enjoyment, difficulty, and self-assessed performance on a 7-point Likert scale (1 = “very false,” 4 = “neutral,” 7 = “very true”). They also reported their familiarity with each study rhythm. These responses captured how different conditions, and auditory feedback influenced the subjective experience of motivation and perceived performance. At the end of the session, participants completed a questionnaire based on the Intrinsic Motivation Inventory [51]. For each condition and auditory feedback type, they rated perceived tension, interest, enjoyment, difficulty, and self-perception of their performance.

2.3 Design and implementation

Rhythm Karaoke is a rhythmic imitation platform that allows users to reproduce a variety of sounds they hear. The system supports the four combinations of a 2 × 2 Melody × Semantic design (monotone notes, melodic notes, monotone speech, and melodic speech), each delivered with tap-synchronised feedback. The monotone notes condition triggers the same C4 piano sample every tap. In the melodic-note condition, the pitch of the sample follows the score of the target tune, so users immediately hear the contour they are trying to imitate. Speech feedback followed the syllabic-control-point strategy of the Voks performative-voice instrument [54, 55, 70], a

gesture-controlled vocal synthesizer in which syllabic timing is defined by P-centers (perceptual centers). Each pre-segmented syllable was tagged with its P-center so that its perceived beat aligned with the user's tap (see Section 2.3.3 "Sound Synthesis" for implementation details). In the monotone speech condition, every tap triggers the next syllable of the target phrase. Before playback, the syllable's detected f_0 contour is flattened to a constant 220 Hz with FMOD's formant-preserving pitch shifter, giving a monotone voice. In the melodic-speech condition, every tap triggers the next sung syllable of the source phrase, preserving its natural pitch contour and timbre.

The sound is generated using a Unity program (see Fig. 2) coupled with the FMOD audio interface. Users can listen to a reference recording to practice imitating the original rhythm of the sound or play with their own interpretation.

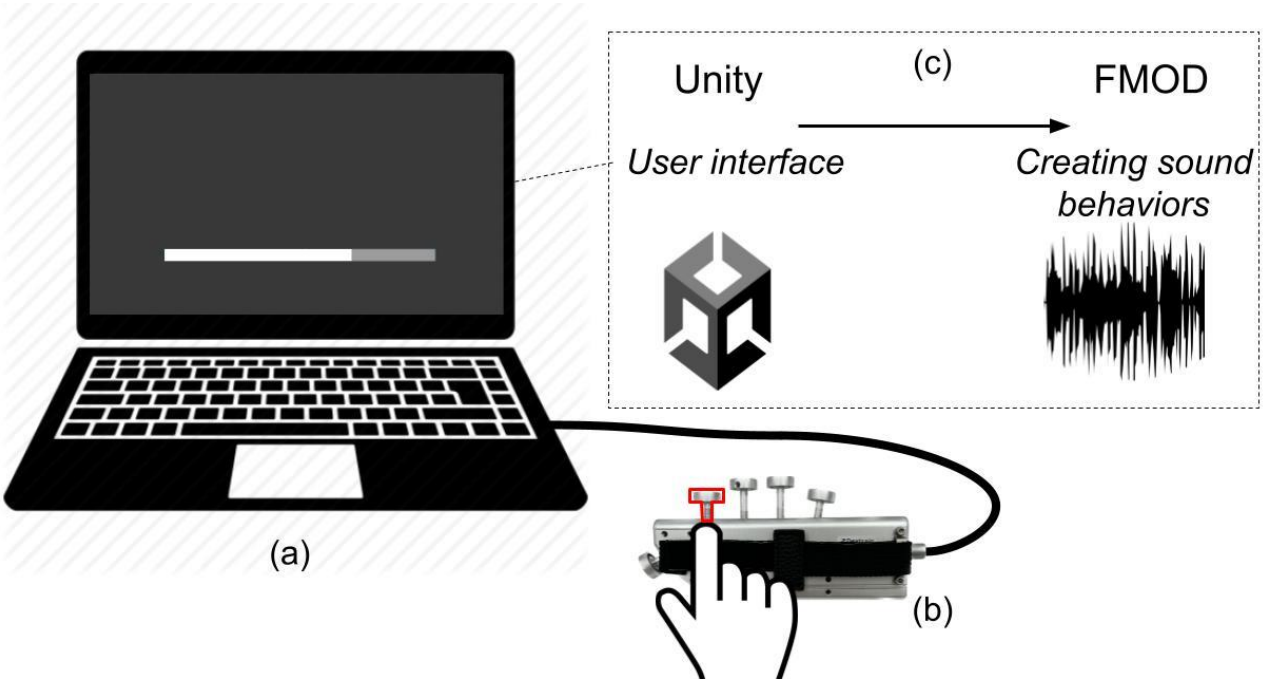


Figure 2. Experimental setup and software chain of the *Rhythm Karaoke* finger-tapping task. (a) Laptop running Unity, which displays the score of the target rhythm as a horizontal progress bar (white = beats already played, grey = upcoming beats) and logs each tap event. (b) The participant taps on the Dextrain Manipulandum (DM) with his/her index finger (highlighted in red). The Dextrain Manipulandum sends a hardware timestamp to the PC with a hardware-to-software latency below 3 ms. (c) Software pipeline: Unity forwards each timestamped tap to FMOD, where event logic converts the tap into (i) the corresponding musical note or (ii) the next speech syllable of the phrase. For speech, syllable boundaries and P-centre tags are pre-computed. FMOD then

applies a formant preserving pitch shift that flattens the syllable's f_0 to 220 Hz in the monotone-speech condition, or it leaves the natural pitch contour unchanged in the melodic-speech condition.

2.3.1 Hardware Interface

The Dextrain Manipulandum (www.dextrain.com) [1], shown in Figure 2, is a device that accurately measures different dimensions of manual dexterity such as force control, rhythmic precision, and the ability to perform individual finger movements.

2.3.2 User Interface

The user interface for Rhythm Karaoke was created in Unity, the software platform for the DM. In our current prototype, a syllable or note is generated each time the user presses the index finger button on the DM. Additionally, a visual cue bar fills up progressively with each press, giving users an idea of their progress through the segment.

2.3.3 Sound Synthesis

The sound in Rhythm Karaoke is generated from real audio files to which we attach precise timestamps. For the note conditions, each timestamp is placed at the acoustic onset of the piano sample, so that a tap fires the note exactly at its attack. For speech we follow the p-centre literature [64, 65]. The p-centre literature defines the perceptual centre of a syllable as the point where it is rhythmically perceived [66, 14], typically where a listener would tap, and it occurs between 20 and 50 ms after the acoustic onset [66, 67]. Following prior implementations that approximate this perceptual lag with a fixed 30 ms delay [68], we apply a 30 ms offset to each syllable nucleus detected by a forced aligner so the timestamp coincides with the perceptual centre heard as the beat of that syllable [56, 57]. This method makes the finger strike feel phase-locked to the speech rhythm.

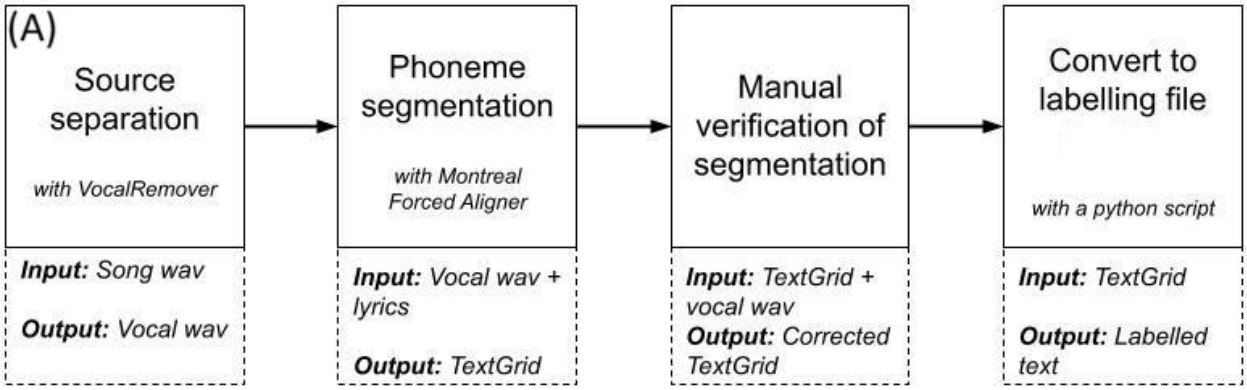
This event list lets users control the rhythm of syllables, notes and speech while preserving the pitch of the original recording [54, 55]. Each button press reads the next timestamp, triggers the corresponding syllable or note in FMOD, and provides immediate auditory feedback. The on-screen progress bar (see Fig. 2) shows how far the user has advanced through the segment.

The playback tempo adapts in real time. After every tap the software measures the current inter-tap interval, compares it with the reference beat period, and sends the results to FMOD. A factor above 1 means the user is tapping faster than the reference, so FMOD compresses the audio in time; a factor below 1 means they are slower, so the

audio is stretched. Because FMOD's time-stretch keeps pitch intact, the sound remains perfectly locked to the user's motion even as they speed up or slow down.

2.3.4 Data Preparation

Before any track can be used in Rhythm Karaoke, speech material must be labelled at the perceptual centres (P-centres) of its syllables, whereas musical notes are timestamped at their acoustic onsets, which coincide with the beat for short percussive sounds. Figure 3A summarises the semi-automatic preparation pipeline. First, the vocal or melodic line is separated from the instrumental accompaniment with Vocal Remover, a free AI-based source-separation tool [33]. Next, for speech stimuli, phoneme segmentation is performed with Montréal Forced Aligner [34], yielding a Praat TextGrid that indicates the start and end times of each phoneme [35]. The TextGrid is then manually inspected in Praat alongside the .wav file, and any segmentation errors are corrected (see Fig. 3B). For melodic-note tracks, the same aligner is run in "note" mode to mark every note onset. Finally, a Python script converts the corrected TextGrid into the JSON cue file required by the playback algorithm (see Fig.3C).



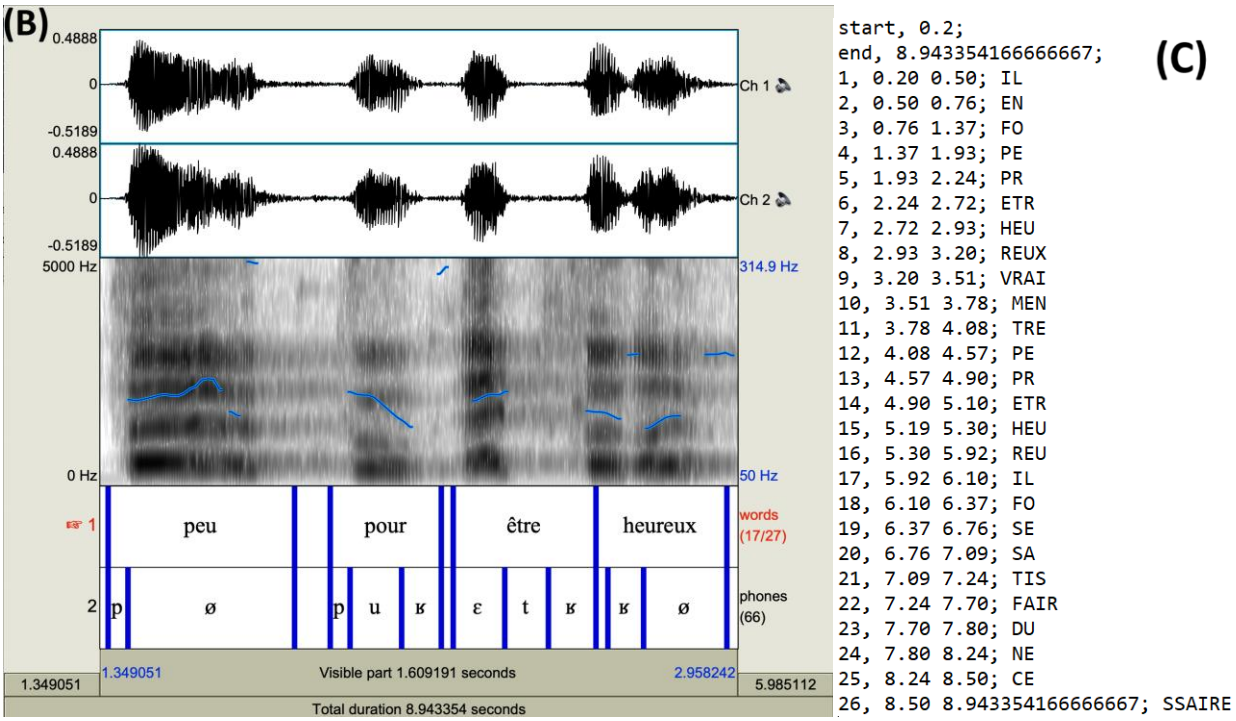
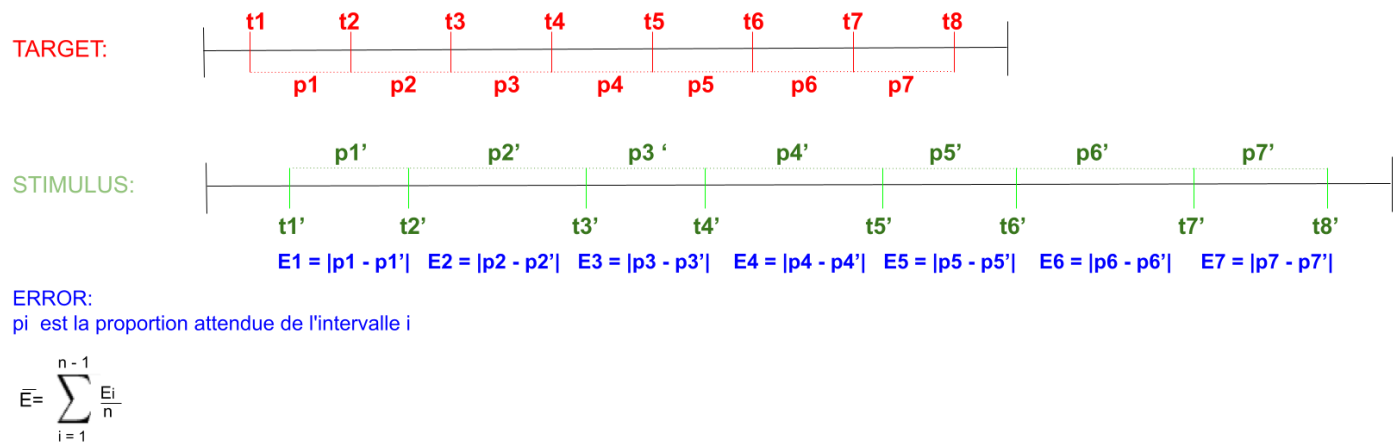


Figure 3. Pipeline for generating time-stamped material that drives Rhythm Karaoke. (A) Pipeline. The vocal line is first isolated with VocalRemover [33], which removes the audio from the audio file and keeps only vocals. Then, phoneme syllable timestamps are generated with the Montréal Forced Aligner [34]. The resulting TextGrid and WAV are bundled into a single upload package. (B) Manual editing stage. The output of the Montréal Forced Aligner is opened in Praat [35] so the operator can drag each boundary to the perceptual centre of its syllable. (C) The corrected TextGrid is then saved alongside the vocal-line WAV.

2.4 Rhythm Error Data Analysis

All tap timings were stored and automatically saved from the Unity interface into .csv files before any analysis. For each task, participants' performance was evaluated by calculating the error between the reference timestamps and those produced by the participants (see Fig.4). In Figure 4, the horizontal line runs from the very first, on left, to the very last tap of the sequence on the right (the full duration of the song/rhythm). Instead of directly comparing the timings, the intervals between timestamps were compared to avoid error accumulation. Moreover, the proportion of intervals (in % of song section length) in the final produced audio was used to evaluate rhythm. This method is suitable for all users, including those with motor deficits, as it does not penalize them for being too slow or too fast, as long as the rhythm is maintained.

At each tap, an error is calculated concerning the corresponding reference timestamp. At the end of each task, the final score is determined by calculating the average error over the entire task. Error scores were averaged per task, removing outliers according to the 1.5 times interquartile range (IQR) method.



$$\bar{E} = \sum_{i=1}^{n-1} \frac{E_i}{n}$$

Fig. 4. Proportional-interval error metric (schematic). A reference timeline (Target, red) and a participant's timeline (Stimulus, green) are shown from the first to the last tap of the rhythm. $t_1 \ldots t_8$ = reference tap timestamps; $t_1' \ldots t_8'$ = participant taps. $p_1 \ldots p_7$ = expected interval proportions; $p_1' \ldots p_7'$ = observed proportions.

2.5 Test-retest Reliability

To evaluate reliability, a second experiment was conducted with 20 participants (7 males and 13 females) with an average age of 25.65 years. This second phase aimed to verify the reliability of the conditions identified as most effective during the first part of the study. Participants performed an initial 15-minute test, followed by a 30-minute break to allow optimal cognitive rest, following recent study recommendations [23, 24]. After the break, participants repeated the same test.

The 12 tasks tested resulted from combining the song section with lowest rhythm error (Balou) and two types of stimuli (melodic note and melodic speech) with and without auditory feedback, each repeated three times.

2.6 Statistical Analysis

The saved data were analysed with Python using the libraries numpy, matplotlib, and pandas. Statistical analysis was conducted with repeated-measures ANOVA, followed by paired t-tests. Before running any ANOVA, the normality of residuals was verified

with Shapiro–Wilk tests and homogeneity of variances with Levene's test, satisfying the assumptions required for parametric inference.

The assumptions of homogeneity of variance and normality of residuals were checked using Levene's and Shapiro-Wilk tests, respectively. A three-way within-subject ANOVA was run with factors Song (3 songs), Melody (2 modes) × Semantic (2 modes) and Feedback (2 modes), plus all interactions. When ANOVAs indicated a significant difference, multiple comparison paired t-tests were conducted to identify specific differences between conditions.

Measurement reliability was assessed using the intraclass correlation coefficient (ICC) model (2,1), suitable for evaluating test-retest reliability [25, 26].

3. RESULTS

3.1 Analysis of Variance (ANOVA): main effects

All participants completed every trial: no technical failures or drop-outs were recorded, indicating good feasibility of the task. ANOVA analysis of reaction times revealed a significant main effect of Time ($F(3, 87) = 3.28$, $p = 0.026$), indicating that participants' reaction times decreased across the session (see Section 3.5 below) ruling out participant fatigue during testing. The ANOVA showed significant effects of Song ($F(2, 667) = 250.57$, $p < 0.001$), Melody ($F(1, 667) = 74.96$, $p < 0.001$) and Semantic content ($F(1, 667) = 44.17$, $p < 0.001$). The effect of Feedback was not significant, ($F(1, 667) = 1.89$, $p = 0.17$). Significant two-way interactions included Melody × Speech, ($F(1, 667) = 74.17$, $p < 0.001$), Song × Melody ($F(2, 667) = 6.82$, $p = 0.001$), Song × Speech ($F(2, 667) = 19.55$, $p < 0.001$) and Speech × Feedback ($F(1, 667) = 7.79$, $p = 0.006$).

3.2 Effect of Song on Rhythmic Error

Paired t-tests compared each song's mean tapping error within subjects ($n = 30$). "Papaoutai" ($M = 2.7$ %, $SD = 0.6$) produced significantly greater error than "Piaf" ($M = 2.0$ %, $SD = 0.4$, $t(29) = 4.80$, $p < 0.001$), and than "Balou" ($M = 1.6$ %, $SD = 0.3$, $t(29) = 7.10$, $p < 0.001$). The error for "Piaf" was also higher than for "Balou," ($t(29) = 2.40$, $p = 0.022$) (see Fig. 5A). The Song × Melody interaction was significant, ($F(2, 667) = 6.82$, $p = 0.001$). Within-song analyses showed that adding a melody reduced error by 1.05 percentage-points for Balou (which relatively represent an error reduction of 42%) ($t(29) = 13.17$, $p < 0.001$), by 0.39 percentage-points for Papaoutai (error reduction of 13%) ($t(29) = 5.06$, $p < 0.001$), and by 0.50 percentage-points for Piaf (error reduction of 23%) ($t(29) = 3.98$, $p < 0.001$) (see Fig. 5B). Similarly, the Song × Speech interaction was significant, ($F(2, 667) = 19.55$, $p < 0.001$): replacing

piano notes with speech reduced error by 1.11 percentage-points for Balou (error reduction of 45%) ($t(29) = 3.68$, $p < 0.001$, by 0.25 percentage-points for Papaoutai (error reduction of 9%) ($t(29) = 1.87$, $p = 0.072$) and by 0.35 percentage-points (error reduction of 16%) for Piaf, ($t(29) = 1.56$, $p = 0.13$) (see Fig. 5C).

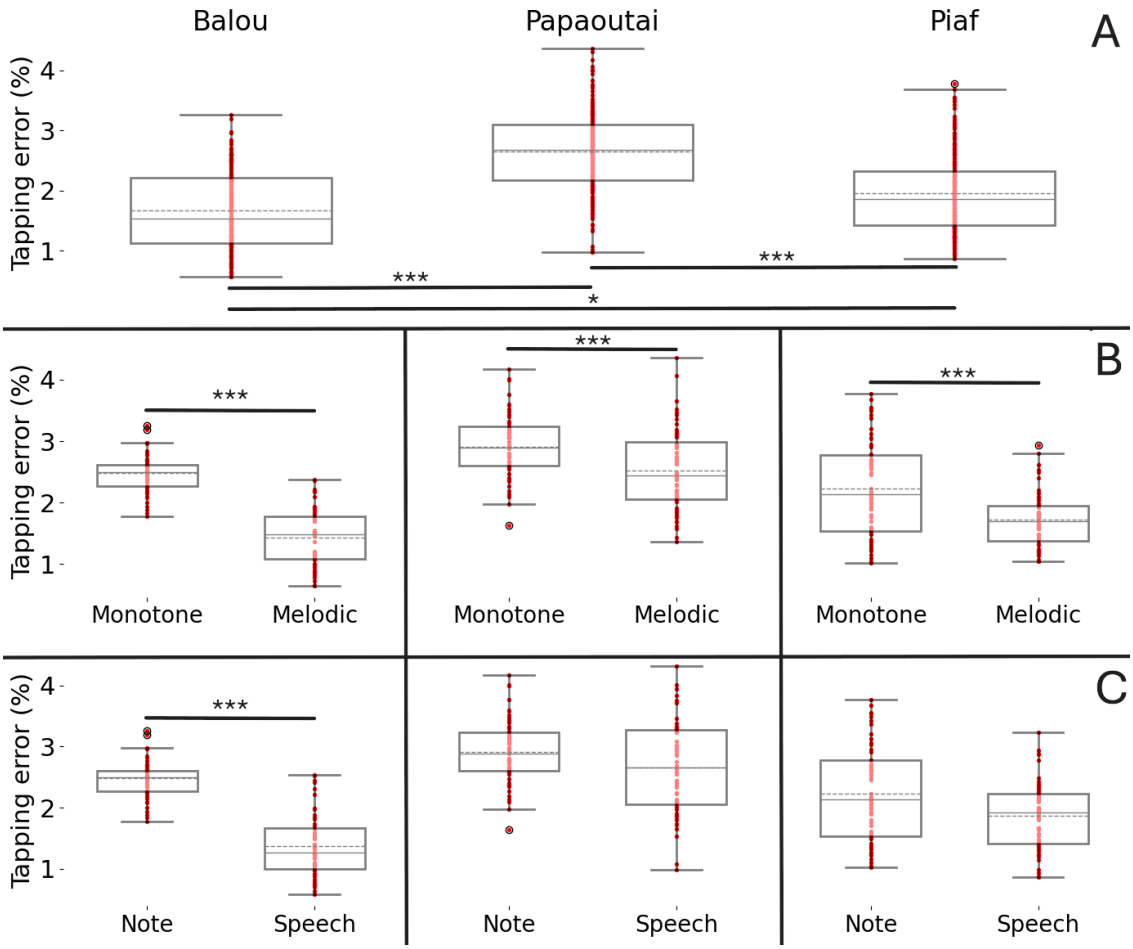


Figure 5 | Effect of song, melody, and semantic content on tapping accuracy ($n = 30$). Each box shows the distribution of percentage errors for every participant. The solid grey line marks the median, the dashed grey line the mean, and red dots are individual scores. A: Error by song. “Papaoutai” ($M = 2.7$ %, $SD = 0.6$) produced larger errors than “Piaf” ($M = 2.0$ %, $SD = 0.4$) and “Balou” ($M = 1.6$ %, $SD = 0.3$).” B: Melodic benefit (Monotone vs Melodic). Adding a melody to the monotone sequences lowered error by 1.05 percentage-points for Balou (error reduction of 42%), 0.39 percentage-points for Papaoutai (error reduction of 13%), and 0.50 percentage-points for Piaf (error reduction of 23%). C: Semantic content benefit (Note vs Speech). Replacing piano notes with spoken syllables reduced error by 1.11 percentage-points for Balou (error reduction of 45%), by 0.25 percentage-points for Papaoutai (error reduction of 9%), and by 0.35 percentage-points for Piaf (error reduction of 16%) (*: $p < 0.05$ for notes, **: $p < 0.01$, ***: $p < 0.001$).

3.3 Main Effects of Melody and Semantic Content

The results also showed that sequences carrying a melody produced lower error by 0.65 percentage-points, which represents a 25 % error reduction over the 2.58 % baseline of the monotone sequences. ($t(29) = 13.55$, $p < 0.001$). Sequences with semantic content (speech) were also more accurate than non-semantic note sequences ($p < 0.001$). When the stimulus consisted of piano notes, adding a melodic contour lowered tapping error by ≈ 0.6 percentage-points ($p < 0.001$). For speech stimuli, the same manipulation had no effect ($p = 0.96$). This difference confirms the Melody × Semantic interaction reported above (see Fig. 5B).

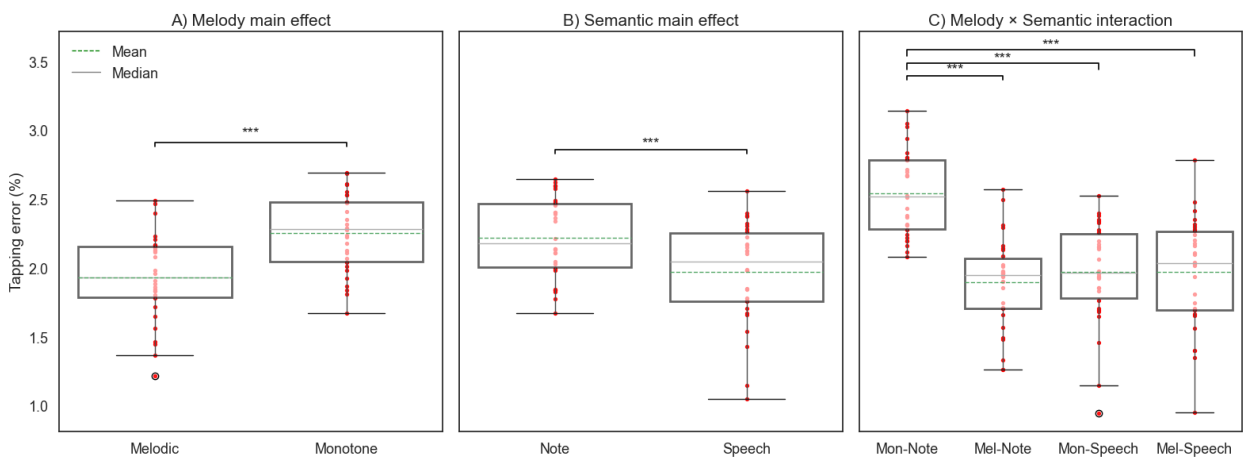


Fig. 6. Tapping-error distributions for the Melody and Semantic factors. (A): The box-plots compare Melodic versus Monotone sequences, illustrating the main effect of the melody. (B): The plots contrast Note versus Speech stimuli, illustrating the main effect of semantic content. (C): The 2 × 2 Melody × Semantic interaction panel shows all four combinations: Mon-Note = monotone note, Mel-Note = melodic note, Mon-Speech = monotone speech, Mel-Speech = melodic speech. For every group, the box spans the inter-quartile range with the median in grey, whiskers extend to 1.5 × IQR, red dots are individual participants, and the green dashed line indicates the group mean (*: $p < 0.05$ for notes, **: $p < 0.01$, ***: $p < 0.001$).

3.4 Interactions with Auditory Feedback

Although the main effect of Feedback was non-significant (see Fig. 7A) ($F(1, 667) = 1.89$, $p = 0.17$), Feedback interacted significantly with Semantic content ($F(1, 667) = 7.79$, $p = 0.006$). In the piano-note conditions, auditory feedback improved reduced error ($t(29) = 2.23$, $p = 0.034$), whereas in the speech conditions feedback modestly increased error ($t(29) = 3.61$, $p < 0.01$) (see Fig. 7B). Feedback showed no interaction with Melody ($F(1, 667) = 0.45$, $p = 0.5$).

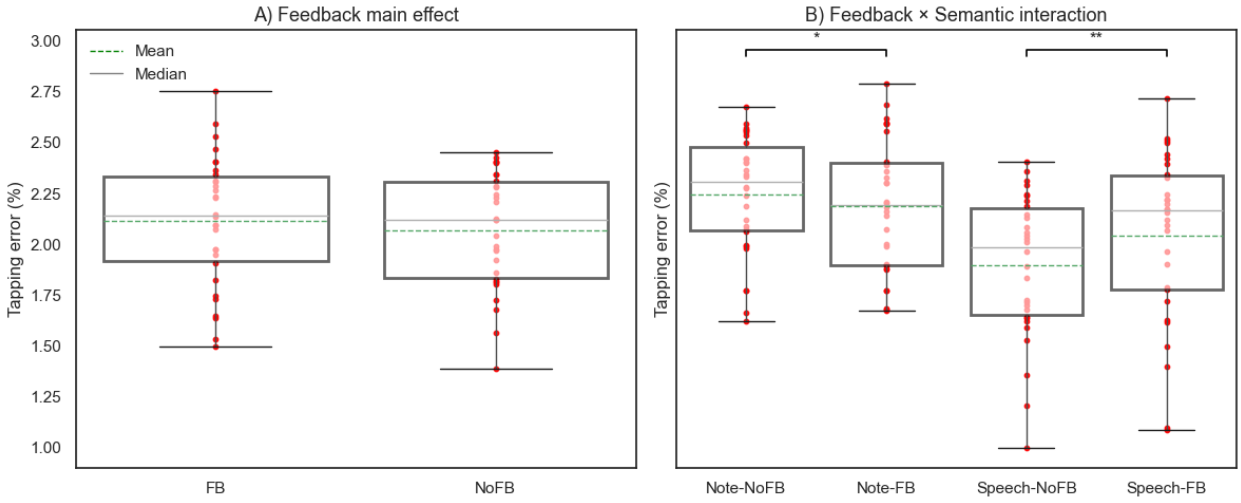


Fig. 7. Tapping-error distributions for auditory Feedback and its interaction with Semantic content. A: Main effect of auditory feedback: FB = trials with feedback, NoFB = trials without feedback, collapsed over Note and Speech stimuli.
B: 2 × 2 Feedback × Semantic interaction. Four box-plots correspond to each combination (Note-NoFB, Note-FB, Speech-NoFB, Speech-FB). Horizontal brackets indicate the feedback effect within each semantic category (*: $p < 0.05$, **: $p < 0.01$). In every plot the box spans the inter-quartile range with the median in grey, whiskers extend to 1.5 × IQR, red dots depict individual participants, and the green dashed line marks the group mean.

3.5 Reaction-Time as a Fatigue Measure

We administered a simple reaction-time task at four time-points (See Fig. 8): T0 (baseline, immediately before the first block of 24 trials), T1 (after 24 trials), T2 (after 48 trials) and T3 (after 72 trials). However, follow-up pairwise t-tests with Bonferroni correction ($\alpha = .008$) showed no significant difference between any two time-points (T0 vs. T1: $p=0.013$; T0 vs. T2: $p=0.011$; T0 vs. T3: $p=0.035$; T1 vs. T2: $p=0.437$; T1 vs. T3: $p=0.331$; T2 vs. T3: $p=0.312$). No fatigue was observed and reaction times tended to decrease slightly early in the session.

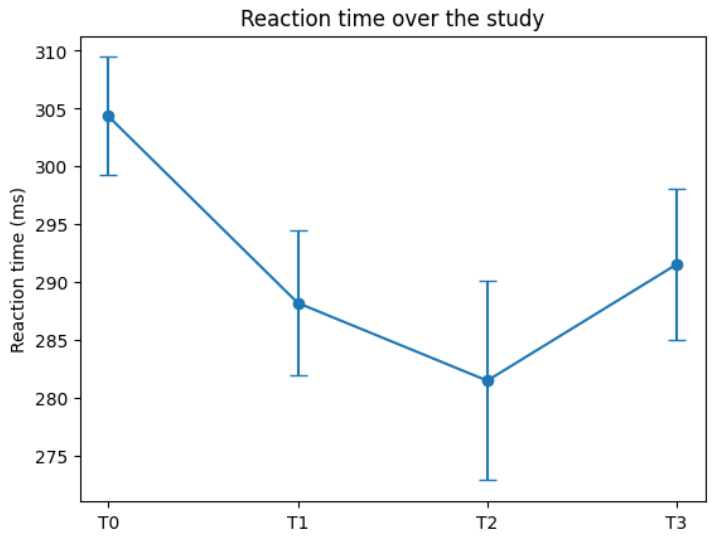


Fig. 8. Simple reaction times (mean ± SEM) measured at four points during the tapping session: T0 = baseline (before the first block of 24 trials), T1 = after 24 trials, T2 = after 48 trials, and T3 = after 72 trials. A one-way ANOVA indicated a significant main effect of Time, $F(3, 72) = 3.28$, $p = .026$, but Bonferroni-corrected pairwise comparisons ($\alpha = .008$) between all time-points were non-significant (all $p > .01$), indicating no reliable fatigue-related change in simple reaction speed.

3.6 Qualitative Evaluation

Participants' Likert scores corroborate the objective findings. A Friedman test revealed a significant main effect of condition for every question on the post-session questionnaire (all $\chi^2(3) \geq 18.2$, $p < 0.001$). Post-hoc Mann-Whitney comparisons (Bonferroni-corrected) consistently placed the two melodic variants ahead of the two monotone ones. In practice, the melodic-note condition yielded the lowest perceived tension and difficulty and the highest interest, enjoyment and self-rated performance; the contrast with monotone speech remained highly significant for all five dimensions ($p= < 0.001$). Melodic speech showed the same favourable pattern, although effect sizes were slightly smaller. Together, these subjective results confirm that participants prefer melodic tasks (see Fig. 9).

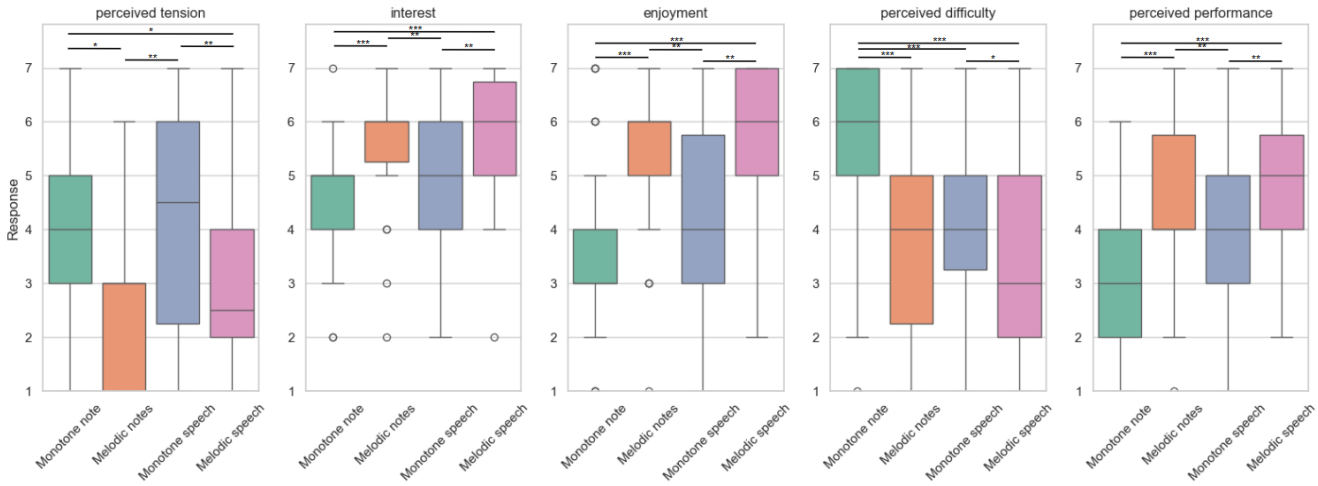


Fig. 9. Qualitative evaluation of conditions by participants. Box-and-strip plots show individual ratings (1 = strongly disagree, 7 = strongly agree) for five questions: perceived tension, interest feeling, enjoyment feeling, perceived difficulty, and perceived performance. Colour coding is consistent across panels: monotone note (green), melodic note (orange), monotone speech (blue), melodic speech (pink). Boxes indicate inter-quartile ranges, black bars mark medians, and black dots represent individual participants (N = 30). The melodic conditions cluster toward lower tension/difficulty and higher interest, enjoyment, and performance (*: $p < 0.05$ for notes, **: $p < 0.01$, ***: $p < 0.001$).

3.7 Measurement Reliability

The reliability results are as follows: the "melodic speech without auditory feedback" condition obtained an intraclass correlation coefficient (ICC [2,1]) of 0.32, while the "melodic speech with auditory feedback" condition obtained an ICC [2,1] of 0.50. For the "melodic note without auditory feedback" condition, the ICC [2,1] was 0.74, and for the "melodic note with auditory feedback" condition, the ICC [2,1] was 0.91. These results indicate that the "melodic note" condition has more reliable measurements than the semantic condition ("melodic speech"). Moreover, although the previous experiment showed that auditory feedback does not significantly influence performance in most cases, it improves measurement reliability.

4. DISCUSSION

We developed a method for rhythmic imitation of finger movements to various song rhythms and studied the effects of different auditory stimuli and providing real-time feedback of performance on rhythmic precision of finger tapping movements in healthy control participants. We also established test-retest reliability in a separate group of healthy participants. The task was feasible, and no effects of fatigue were found during

the testing session. As hypothesized, participants showed a benefit, in terms of reduced rhythmic error, when the auditory stimuli were melodic and contained semantic content (speech). Auditory feedback enhanced rhythmic precision in notes but had a detrimental effect in speech conditions. Melodic and speech stimuli were also the conditions with highest subjective ratings of interest and motivation. Tasks showed moderate-excellent reliability with better reliability found in melodic note compared to melodic speech conditions. The findings will be discussed in turn below.

4.2 Melodic and semantic auditory stimuli improve rhythmic tapping performance

Our results revealed significant effects of song rhythms and conditions on participants' performance. Notably, the rhythm "Papaoutai" by Stromae proved the most difficult, while "Balou" from The Jungle Book was the easiest. This can be explained by the fact that "Papaoutai" is a fast and irregular rhythm, increasing cognitive load and resulting in decreased performance in a motor task [19]. Moreover, the rhythm of "Piaf" involves prolonged silences, which can also increase the cognitive load by disrupting rhythmic movement continuity [20, 13]. Conversely, "Balou " is a slower and more regular rhythm, making the task easier for participants.

The "monotone note" condition was the most difficult, consistent with the work of Tierney and Kraus [18], which shows that monotone rhythmic stimuli are more complex to process due to the absence of melodic or semantic variations to guide participants [28].

Melodic stimuli improved rhythmic precision compared to monotone stimuli, supporting our initial hypotheses. These results align with the research of Thaut and Kenyon [17], which shows that melodic stimuli facilitate rhythmic and motor processing by providing additional auditory cues. Furthermore, participants preferred melodic conditions, reflecting the conclusions of Altenmüller and Schlaug [4] on the positive impact of music on motivation and engagement.

Our results highlight the importance of melody and speech in facilitating rhythmic tasks, as reported in previous studies [21, 27, 12]. For example, Zatorre and Salimpoor [21] showed that melodic stimuli activate neural networks associated with pleasure and motivation, which may explain why our participants preferred these conditions. Additionally, Christiner et al. [7] demonstrated that melodies and languages are perceived as more memorable and accessible [16], thus facilitating rhythmic imitation.

Our results also indicate that tapping was more accurate when guided by spoken syllables than by monotone note cues. Similar effects have been observed when finger-tapping is paced by syllables or spoken digits rather than metronome clicks of the

same tempo [78, 79]. Speech carries a prosodic structure, where the amplitude envelope and stressed syllables provide temporal anchors. Luo & Lu [77] review shows that speech rhythms recruit the dorsal audio-motor pathway and strengthen auditory–motor coupling. In our task, the syllables likely provided both temporal and semantic cues that facilitated motor timing and improved rhythmic precision.

However, unlike some studies suggesting that auditory feedback systematically improves performance [56, 57], we observed a complex interaction between conditions and auditory feedback. In some conditions, notably semantic stimuli, auditory feedback disturbed participants' performance. This could be attributed to the novelty of the performative, synthesised speech used for feedback and to the quality of that synthesis [58, 59, 60], as well as the greater involvement of cognitive resources needed to process speech, as suggested by Hardy et al. [8]. The reliability results also showed better test-retest results in melodic notes versus melodic speech conditions. This may also be explained by semantic condition engaging more varying cognitive resources that sometimes can be disturbing. Our results extend previous results showing excellent reliability in non-melodic note tapping on a tablet-based application [71].

4.3 Limitations

Our findings should be interpreted by taking in consideration several factors that may have shaped participants’ performance. For example, participants' familiarity with the rhythms used could have influenced their performance [5]. However, except of 5 participants that didn’t know the song of “Balou”, the population tested all knew the songs tested as indicated in post-testing comments. Moreover, the variability of results between conditions could be due to individual differences in rhythm perception and production, as suggested by the work of Repp [14]. Here we present group comparisons using ANOVA analysis which identifies factors that influence whole group in systematic ways and does not allow identification of sub-groups. Finally, the interface used, being new to participants, may have affected the observed results [61], but practise allowed participants to get familiarized before testing.

Our study has several limitations. The sample size, although justified by a power analysis, remains relatively small and may limit the generalizability of the results. Additionally, the lack of strict control over the starting and ending conditions of the trials could introduce biases, particularly regarding participants' familiarity with the rhythms [9]. Finally, auditory-feedback quality varied because the system time stretches every note or syllable to stay in phase to each tap. Tapping the sample too slowly or too quickly produces micro-stretch artefacts which can make the feedback sound less natural and disturb timing. Even if some studies suggest that auditory feedback improves performance [56, 57], we observed a complex interaction between conditions

and auditory feedback. One explanation is that our speech feedback pipeline still introduces small but audible timing and timbral artefacts. In particular, the fixed 30 ms offset chosen to approximate each P-centre may be too coarse, since true P-centres vary with syllable structure and vowel quality [68]. Together with the 50–100 ms output latency that is typical of real-time FMOD playback on consumer hardware and the micro-stretching applied to keep pitch constant, these misalignments can make the feedback sound less natural and can therefore perturb user tasks. Tapping accuracy could improve if each syllable's P-centre were estimated from its own acoustic landmarks rather than assigned a uniform 30 ms delay. Future versions should therefore estimate P-centres on a syllable-by-syllable basis or let the offset adapt online. Recent work on performative speech synthesis shows that tapping accuracy increases when each syllable is controlled with two-time anchors (vowel nucleus and release) rather than a single one, and that the choice of acoustic anchor (vowel nucleus versus measured P-centre) also modulates precision [69, 70].

4.4 Implications of the Results

These findings have implications for both motor rehabilitation and the design of interactive musical interfaces. The improvement in rhythmic precision due to melodic and semantic stimuli suggests that these types of stimuli could be integrated into rehabilitation programs to maximize patient engagement and motivation [15]. Moreover, participants' preference for melodic conditions highlights the importance of designing musical interfaces that incorporate melodic elements to enhance user experience [6].

Future research should include larger samples to improve the robustness of the results. It would also be pertinent to explore the impact of musical familiarity on rhythmic performance and to provide high-quality auditory feedback. Additionally, it would be interesting to conduct studies on individuals with motor deficits, such as stroke patients, to examine how melodic and semantic stimuli influence manual dexterity rehabilitation and functional recovery. Finally, further studies should explore the neurophysiological mechanisms underlying the observed effects, notably using brain imaging techniques to analyze changes in neural activity associated with rhythmic imitation.

5. CONCLUSION

This study presents a novel method for rhythmic production of finger tapping movements to a variety of auditory stimuli. Our results show that melodic and semantic stimuli significantly improve rhythmic precision compared to monotone stimuli, supporting the study's initial hypotheses. Additionally, participants expressed a marked preference for melodic conditions, highlighting the importance of these elements for engagement and motivation.

Complex rhythms, such as "Papaoutai" by Stromae, proved more difficult to imitate, while simpler rhythms like "Balou" facilitated participants' performance. The presence of auditory feedback showed a complex interaction with experimental conditions, suggesting that, while beneficial in note conditions, it can disrupt performance in speech conditions.

These results have significant implications for motor rehabilitation and the design of interactive musical interfaces. Integrating melodic and semantic stimuli into rehabilitation programs could maximize patient engagement and motivation [22, 29, 30, 31], thus improving therapeutic outcomes. Moreover, the reliability of rhythmic precision measurements in melodic conditions with auditory feedback highlights the importance of these parameters for accurate and repeatable evaluation.

Future research will explore the impact of musical familiarity on rhythmic performance in neurological conditions, such as stroke. Finally, using brain imaging techniques could provide a deeper understanding of the neurophysiological mechanisms underlying the observed effects, paving the way for even more effective rehabilitation approaches.

DECLARATIONS

Ethics approval and consent to participate:

The study was performed according to the declaration of Helsinki. This study received the approval of the Ethics Committee of the Université Paris Cité (CER U-Paris Cité N◦ IRB: IORG0010120-2025-33).

Consent for publication:

Not applicable

Availability of data and materials:

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests:

PGL has patented the method for multidimensional measurement of manual dexterity (WO2016184935A3) and is founding members of start-up company Dextrain (started in 2021) PGL reports no financial gain in advising the company. The other authors report no financial interests or potential conflicts of interest.

Funding:

PhD funding for PPA was provided by joint Devinci-Dextrain financial support. PGL is supported by the HapticS project grant (French National Research Agency, ANR-22-CE19-0011).

Authors' contributions:

PPA, XX and PGL designed the study; PPA developed rhythmic dexterity platform; PPA performed data acquisition and analysis; PPA, XX and PGL wrote the manuscript. All authors read and approved of the final manuscript.

Acknowledgements:

We express our appreciation to the participants of the study. We thank Christophe d'Alessandro and his team at the Lutheries-Acoustique-Musique (LAM) group, Institut Jean Le Rond d'Alembert – CNRS / Sorbonne Université, for their valuable discussions and technical support in developing the Rhythm Karaoke interface.

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